Platform Sponsor Declaration: Replit Infrastructure Integration

1. Purpose & Architectural Role

In our platform architecture, Replit serves as the unified, full-stack hosting and operational environment for the Adept web application and API services. Rather than managing disparate hosting providers for our frontend, backend, and database, we consolidated our entire live infrastructure onto Replit Core to ensure seamless deployment and tight integration between our Next.js UI, FastAPI microservices, and persistent data storage.

2. Exact Workload Executed

- **Task:** End-to-end continuous deployment, live application hosting, and persistent relational database management.
- **Infrastructure Utilized:**
    - **Application Hosting:** Autoscale infrastructure deployed in the Asia region (2 vCPU / 4 GiB RAM).
    - **Database:** Replit Integrated PostgreSQL Database (Development and Production instances).
- **Execution Metrics:**
    - Live public URL assignment (adept-my.replit.app).
    - Continuous integration and publishing pipeline utilizing Replit Deploy.
    - Active cost and resource monitoring (e.g., tracking a $6.38 usage spend over the current period).

3. Downstream System Impact

Replit’s integrated PostgreSQL database securely stores our entire application state. This includes 140 canonical skills, 160 role-skill mappings, 98 verified courses, and our crucial pgvector embeddings (skill_embeddings table). By hosting the database directly within the Replit ecosystem, we achieve low-latency data retrieval for our Fast/Moderate scenario models and semantic matching engine, ensuring a highly responsive user experience.

4. Compliance with Hackathon Boundaries

- **Centralized and Transparent Hosting:** Utilizing Replit for our deployment demonstrates a clean, auditable architecture. The integration of the Replit AI Agent for development assistance (as seen in the workspace) aligns with the hackathon's focus on utilizing modern AI tooling responsibly.
- **Reliable Prototype Delivery:** By leveraging Replit's robust Autoscale infrastructure and integrated production database, we guarantee a stable, publicly accessible live prototype (adept-my.replit.app) for the judges, eliminating the risk of local

environment configuration errors or database connection timeouts during the live demo.

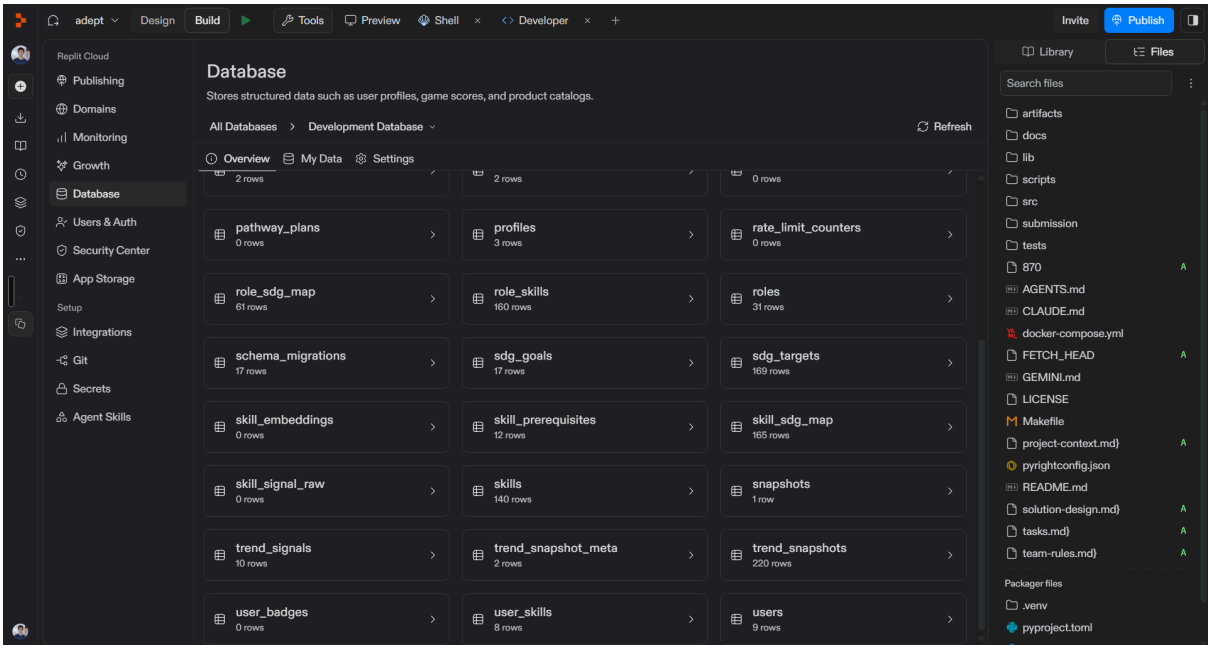


Fig. 1. Proof of Replit Database Utilization (Darren)

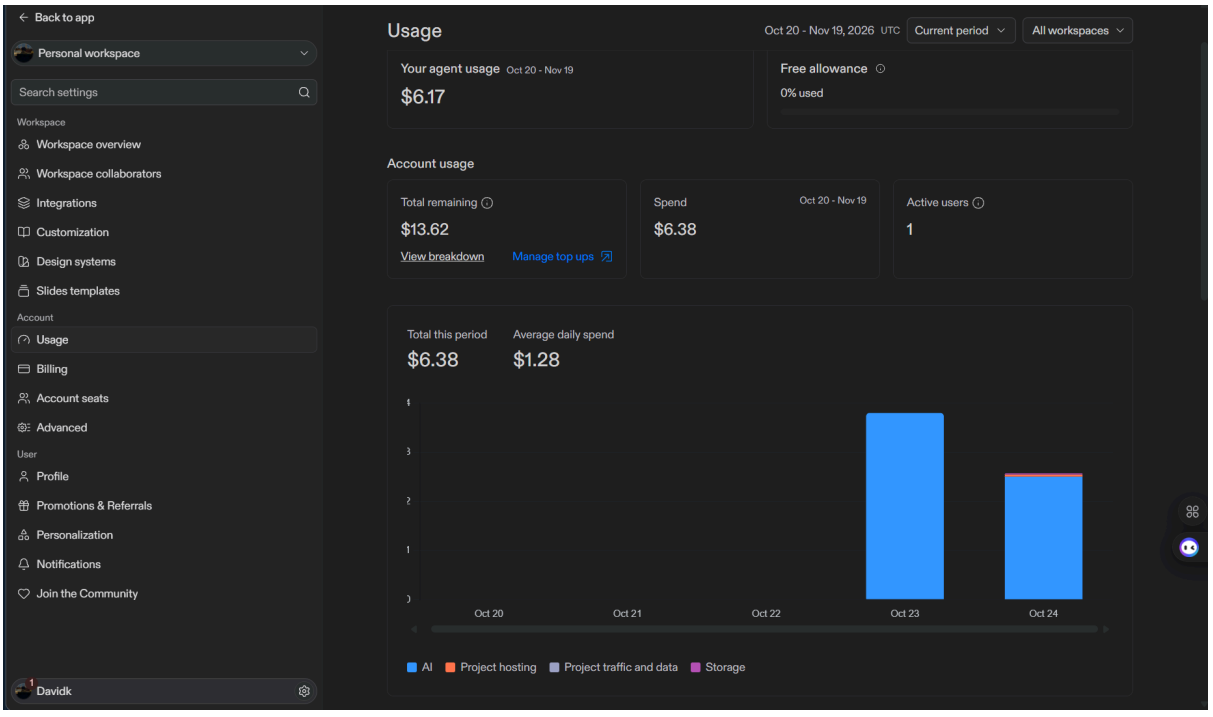


Fig. 2. Proof of Remaining Account Usage Credit (David)

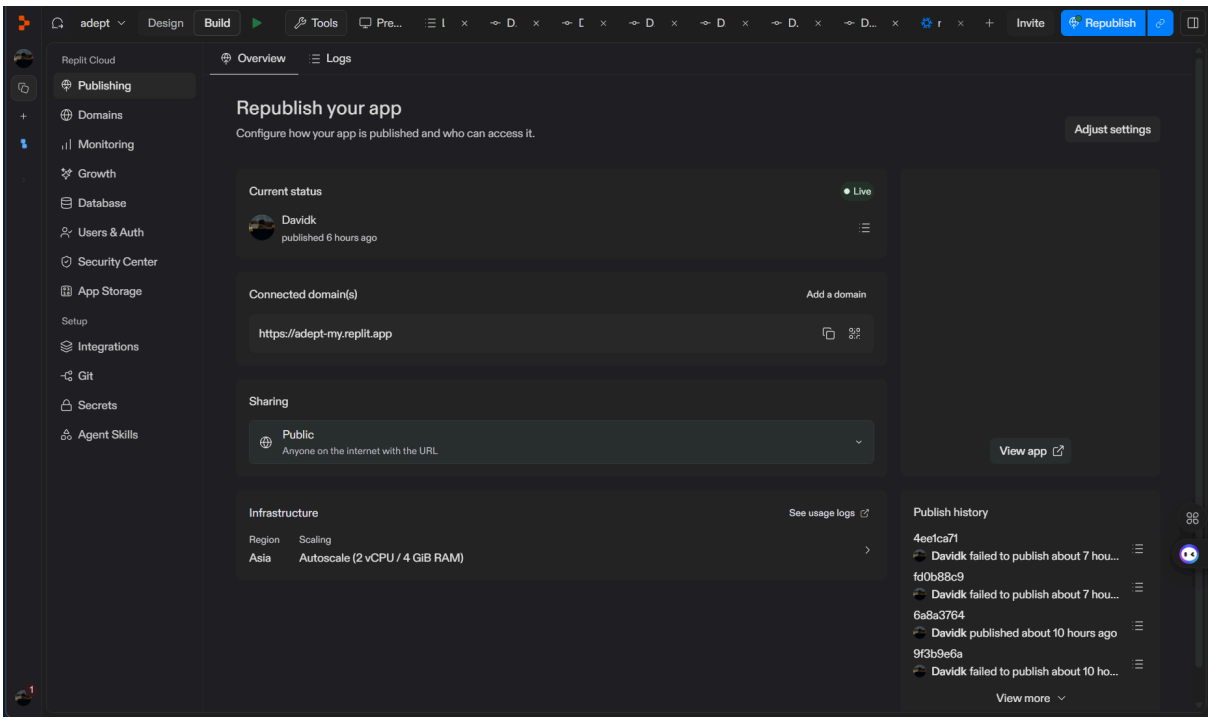


Fig. 3. Proof of Publishing an App via Replit (David)

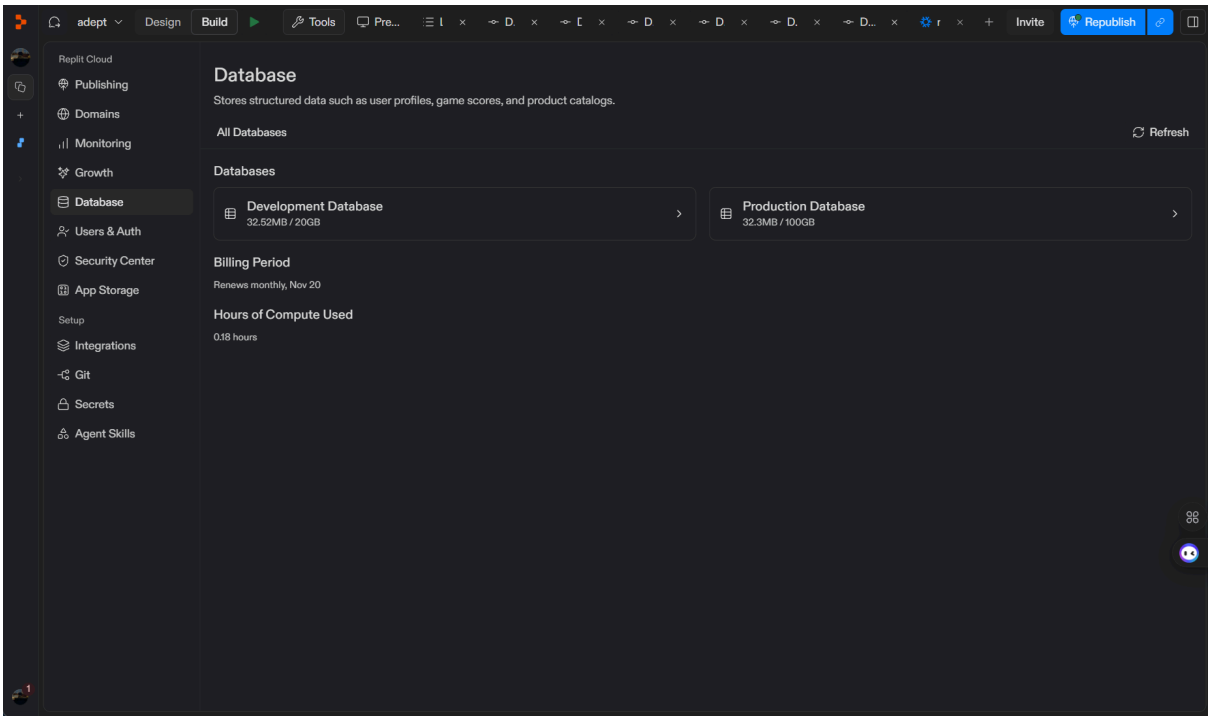


Fig. 4. Proof of Utilizing Replit Integrated Database (David)